

Title: Decomposing Place Names: Evidence from the Hebrew Construct State

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Abstract

This paper claims that place-denoting proper names (i.e., toponyms) are structurally decomposable. Evidence for this claim comes from Hebrew toponyms displaying noun-noun concatenation (e.g., *Tel-Aviv*). I show that such toponyms share properties with the Hebrew Construct State construction. Using Borer's (2013) typology of Constructs, I investigate how these toponyms pattern with Construct subclasses using a range of diagnostics. Accordingly, it is demonstrated that such toponyms are decomposable in two distinct ways. Specifically, they instantiate two subclasses: (a) toponymic Modificational-Constructs, a grammatically transparent class, and (b) toponymic Compounds, a grammatically opaque class. Each subclass corresponds to a distinct structural configuration. By structural parsimony, I propose that these configurations reduce to structures independently available for Constructs in the language. I further claim that the non-compositional meaning of toponymic Compounds corresponds to geo-ontological categories (e.g., [CITY], [VILLAGE], [HILL]; cf. Smith & Mark 2001), suggesting a principled mapping between geographical concepts and syntactic structure.

Keywords: toponyms; place names; Hebrew; construct state nominals; compounding; proper names; Semitic languages; geo-ontological categories

1. Introduction

Place names (i.e., toponyms) in Hebrew often occur as noun-noun concatenations, as illustrated by the following examples:¹

- | | |
|---|--|
| (1) a. <i>yam</i> <i>ha-melax</i>
sea.CS the-salt
(Dead Sea) | d. <i>beyt</i> <i>šemeš</i>
house.CS sun
(Bet Shemesh) |
| b. <i>gva'ot</i> <i>ha-šokolad</i>
hills.CS the-chocolate
(Chocolate Hills) | e. <i>petax</i> <i>tikva</i>
opening.CS hope
(Petah Tikva) |
| c. <i>arcot</i> <i>ha-brit</i>
states.CS the-union
(The US) | f. <i>tel</i> <i>aviv</i>
mound.CS spring
(Tel Aviv) |

Such N_1+N_2 toponyms display properties commonly associated with the so-called construct state construction (henceforth, simply Construct), as exemplified in (2a); a genitive construction well known in Semitic and from studies on Hebrew (e.g., Borer 2013; Danon 2025; Ritter 1991; Shlonsky 2004; Siloni 2001).² Hebrew Constructs are typically identified by the following cluster of properties (see comparison between toponymic and non-toponymic Constructs in parentheses): (a) N_1+N_2 combinations – the noun N_1 heading the construction occurs in initial (leftmost) position (e.g., *beyt* in (2a); *yam* in (1a)) and is followed by a noun phrase (i.e., the non-head N_2 : e.g., *ha-mora* in (2a); *ha-melax* in (1a)), understood as the genitive member of the head

¹ Abbreviations: CS = construct state, FS = free state, F = feminine, M = masculine, SG = singular, PL = plural, PRS = present, ADJ = adjectivizing suffix, LM = linking-morpheme, N_1+N_2 = noun-noun concatenations.

² For Constructs in other Semitic languages, see e.g., Fassi Fehri (1993) and Benmamoun (2000) for Arabic, and Kramer (2010) for Amharic. It should further be noted that many Hebrew $N+N$ toponyms emerged through processes of Hebraization of earlier place names in Palestinian Arabic (see Benvenisti 2000). The historical and political dimensions of these naming practices fall beyond the scope of the present paper.

noun N₁; (b) *Strict adjacency* – the head noun N₁ must directly precede the non-head N₂ without the intervention of any overt material such as the dummy preposition *šel* ‘of’ (e.g., cf. (2a) vs. *beyt* (**šel*) *ha-mora* or (1c) vs. *arcot* (**šel*) *ha-brit*); (c) *The article constraint* – the definite article (if present) usually cannot appear on the head N₁ (e.g., cf. (2a) vs. (**ha-*)*beyt* (*ha-*)*mora*; (1a) vs. (**ha-*)*yam* (*ha-*)*melax*)), rather it attaches to the non-head rendering the entire construction definite (cf. (1a-c) vs. (2a))³; (d) *Morpho-phonological peculiarities* – the head noun N₁ often undergoes morpho-phonological changes (e.g., cf. (1d) and (2): *bayit* (free-state) → *beyt* (construct-state)).⁴

(2) a. Construct state:

beyt ha-mora
house.CS the-teacher.F.SG
‘The teacher’s house’

b. Free state:

ha-bayit šel ha-mora
the-house.FS of the-teacher.F.SG
‘The house of the teacher’

Despite occasional references to N₁+N₂ toponymic Constructs such as those in (1), they have largely remained a peripheral topic in the generative literature. Instead, attention has been directed toward distributional asymmetries of toponyms vs. personal names in Constructs more generally. For example, unlike personal names, toponyms, including N+N toponymic Constructs

³ In colloquial Hebrew, the definite article may surface on the head noun in certain subclasses of Construct States; see Borer (2013) for discussion.

⁴ More notations: (*X) marks an ungrammatical form, and (X) marks an optional element. Note that the constituent referred to here as the non-head has also been termed the annex (e.g., Rothstein 2012) or the dependent (e.g., Preminger 2020) in the literature. Following Borer (2013) and Siloni (2001), I adopt the term non-head throughout.

in (1), have been shown to occur freely as non-heads in Constructs in Modern Hebrew (cf. Rothstein 2017, 2012; Danon 2017; Jeffay & Rothstein 2022), as illustrated in (3).⁵

(3) a. *#yaldey ron*
children.CS Ron
Intended: ‘The children of Ron’

b. *yaldey tel aviv*
children.CS Tel Aviv
‘The children of Tel-Aviv’

This asymmetry has been taken as evidence that toponyms and personal names in Modern Hebrew belong to distinct grammatical classes (see, e.g., Rothstein 2012, 2017). However, the internal composition of N_1+N_2 toponymic Constructs has remained overlooked.

Based on the observation that $N+N$ toponymic Constructs share core properties with genuine non-toponymic Constructs in Hebrew, the goal of this paper is therefore to address the following two questions: (a) What is the internal syntactic composition of toponymic Constructs, if any? and (b) How do such toponyms fit within the broader typology of Hebrew Constructs?

A prerequisite for answering these questions is an understanding of how Hebrew Constructs generally pattern in the language. Previous research has shown that Hebrew Constructs do not constitute a uniform class. Rather, on the basis of a range of diagnostics, Borer (2013) establishes a typology comprising three distinct Construct classes: (a) Referential-constructs (R-Constructs), (b) Modificational-constructs (M-Constructs), and (c) Compound Constructs.⁶ The first two classes are compositional insofar as the meaning of the whole is predictable from the meanings

⁵ Interestingly, Biblical Hebrew permits personal names to occur as non-heads in Constructs. See Jeffay and Rothstein (2022) for further discussion.

⁶ This typology of Constructs has subsequently been extended to other languages. See, for example, De Belder (2022) for Dutch.

of the nominal subparts. They differ, however, with respect to the status of the non-head N_2 : in R-Constructs (e.g., (4a)), N_2 is a full referential constituent (i.e., a DP), whereas in M-Constructs (e.g., (4b)), N_2 functions as a property modifier (i.e., an NP predicate). Unlike the first two classes, the third class, Compound Constructs, is non-compositional insofar as the meaning of the whole is not predictable from the meanings of the nominal subparts (e.g., (4c)).⁷

(4) a. R-Constructs:	b. M-Constructs:	c. Compound Constructs:
<i>beyt ha-mora</i>	<i>beyt ha-‘ec</i>	<i>beyt ha-sefer</i>
house.CS the-teacher.F.SG	house.CS the-wood.M.SG	house.CS the-book.M.SG
‘The teacher’s house’	‘The wooden house’	‘The school’
(Compositional)	(Compositional)	(Non-compositional)

Against this background, the question arises as to whether N+N toponymic Constructs instantiate one of the established Construct classes above and, if so, what this may reveal about their internal structure. The typology in (4) provides a natural basis for addressing such questions. Specifically, if N+N toponymic Constructs are genuine Constructs, they are predicted to exhibit the diagnostic properties associated with one or more of the established Construct classes. In Section 2 it is demonstrated that this prediction is borne out. The extent to which such toponyms pattern with R-Constructs, M-Constructs, and Compound Constructs therefore offers a principled means of determining their grammatical status and internal composition.

Given the widely accepted view that Constructs are generated in the syntax (cf. Borer 2013; Ritter 1991; Shlonsky 2004; Sichel 2002), the finding that N+N toponymic Constructs pattern with established Construct classes provides direct evidence that they possess complex internal syntactic structure. This conclusion stands in contrast to the view that proper names, including toponyms, are syntactically and semantically atomic linguistic objects lacking complex internal

⁷ I adopt Borer’s (2011) label R-constructs for referential constructs; these correspond to what Borer (2013) later terms I(individual)-Constructs following Munn (1995).

structure (i.e., non-branching DPs; Abney 1987) and interpreted merely as rigid designators or constants (Kripke 1980; Soames 2002).

Drawing on evidence from Hebrew, this paper advances the claim that toponyms are structurally decomposable, thereby supporting the view that proper names are in actuality syntactically and semantically complex linguistic objects (cf. Matushansky 2015; Fara 2015). Concretely, it is demonstrated that N+N toponymic Constructs are decomposable in two distinct ways. Applying the diagnostics, I show that N+N toponymic Constructs display only partial convergence with non-toponymic Constructs. Specifically, the diagnostics reveal that N+N toponymic Constructs do not constitute a uniform class but instead instantiate two distinct subclasses: (a) toponymic M-Constructs and (b) toponymic Compounds, to the exclusion of toponymic R-Constructs. The diagnostics further indicate that toponymic M-Constructs have phrasal status, whereas toponymic compounds have wordhood status.

The diagnostic differences between the two toponymic subclasses further suggest that each subclass corresponds to a distinct underlying structural configuration. In the interest of structural parsimony, I propose that such structural configurations are reducible to structures that are independently available in the language for non-toponymic Constructs. Concretely, toponymic M-Constructs are assigned the same compositional structure as their non-toponymic counterparts, in which the non-head N_2 functions as a property modifier, namely an NP predicate. In contrast, toponymic Compounds require a structure that would account for their non-compositional wordhood status. Following Borer (2013), it is proposed that it is the process of head incorporation that gives rise to the attested wordhood status of toponymic compounds.

In the spirit of the Exo-Skeletal Model (henceforth XS-model; Borer 2005, 2013), I further propose that the non-compositional meaning assigned post-syntactically to the incorporated structure of toponymic Compounds corresponds to atomic content units associated with geo-ontological categories, broadly corresponding to concepts such as [CITY], [VILLAGE], [HILL], and so on (cf. Smith & Mark 2001). In this respect, the mapping between geographical concepts and syntactic structures associated with non-compositional meaning is shown to be principled.

The paper is organized as follows: §2 provides the diagnostic and empirical basis for reducing toponymic N+N Constructs in Hebrew to the structures of M-constructs and compound constructs. §3 presents the formal analysis and discusses in more details the internal syntax and semantics of Hebrew N+N toponyms. §4 summarizes the key findings and outlines directions for further research.

2. Diagnosing Hebrew toponymic Constructs

This section introduces a series of diagnostic tests to determine the grammatical status of toponymic Constructs such as those in (1). Through a comparison between toponymic and non-toponymic Constructs in Hebrew, it is demonstrated that although the two share several properties, they diverge on a number of crucial diagnostics. Specifically, while toponymic and non-toponymic Constructs share the properties listed under *Similarities* in Table 1 (properties (a) and (b) discussed in §1, and property (c) discussed in §2.1), toponymic Constructs do not display some core properties characteristic of R-Constructs. Rather, they pattern with M-Constructs and Compound Constructs. The diagnostics therefore point toward a division of toponymic Constructs into two distinct subclasses: (i) toponymic M-Constructs and (ii) toponymic Compounds.

The findings, which later inform the formal syntactic analysis proposed for toponymic Constructs, are summarized in Table 1. The grey cells highlight the partial convergence of toponymic Constructs within the broader typology of Hebrew Constructs.

Table 1: Typology of toponymic Constructs in Hebrew

		Constructs		
		Toponymic Constructs		
		Compound Constructs	M- Constructs	R- Constructs
<i>Similarities</i>	a. Strict adjacency	✓	✓	✓
	b. Definite article on the non-head	✓	✓	✓
	c. Stress-assignment domain	✓	✓	✓
<i>Differences</i>	a. Semantic headedness (compositionality)	✗	✓	✓
	b. Pronominal reference to the head	✗	✓	✓
	c. Pronominal reference to the non-head	✗	✗	✓
	d. non-head modification	✗	✗	✓
	e. Wordhood or phrasal status	wordhood	phrasal	phrasal

In what follows, I first complete the discussion of the similarities as sketched in Table 1 by looking at the stress-assignment domain (§2.1). I then turn to the diagnostics that distinguish

among the Construct classes: semantic headedness (§2.2), pronominal anaphora (§2.3), non-head modification (§2.4), and wordhood vs. phrasal status (§2.5).⁸

2.1. Similarities

Hebrew (non-)compositional Constructs form single prosodic units (cf. Borer 2013; Faust 2014; Siloni 2001; see also references cited therein). The examples below suggest that the same is true for toponymic Constructs, that is, they exhibit regular morpho-phonological behavior that is expected of Constructs in Hebrew. Concretely, both non-toponymic and toponymic Constructs have only one primary stress falling on the non-head (cf. 5-8). This, in turn, triggers a variety of morpho-phonological changes on the head. For example, free-state (i.e., non-construct) feminine singular forms ending in *-a* instantiate a distinct form by the insertion of /t/ (cf. 5-6). In the same vein, free-state forms with the plural suffix *-im* undergo a morpho-phonological change into a distinct bound form (i.e., the ending /ey/, cf. 7-8).⁹¹⁰

(5) Non-toponymic

a. <i>šmira-t</i>	<i>yeladím</i>	cf. <i>šmira</i>
guarding.F.SG-CS children		guarding.F.SG
‘guarding children’ (compositional)		‘guarding’

b. <i>šmira-t</i>	<i>sáf</i>
guarding.F.SG-CS threshold	

⁸ The diagnostics listed under differences in Table 1 are drawn from Borer (2013) and were selected based on their applicability to toponymic Constructs. Other diagnostics discussed there, such as the displacement of the definite article, non-head coordination, and definiteness spreading, are excluded because the strong namehood status of toponyms tends to neutralize the relevant contrasts. The analysis therefore relies on diagnostics that remain informative in this domain.

⁹ Examples (5) and (7) are adapted from Borer (2013: 211).

¹⁰ Note, however, that some exceptions do exist. Noam Faust (p.c.) notes that in some place names primary stress may shift to the head (e.g., *bat yám* → *bát yam* [city name]). I will leave a more detailed investigation of this observation for future research.

‘gate-keeping’ (non-compositional)

(6) Toponymic

a. <i>rama-t</i>	<i>gán</i>	cf. <i>rama</i>
plateau.F.SG-CS	garden	plateau.F.SG
(city name)		‘plateau’

b. <i>giv'a-t</i>	<i>ha-‘irus-ím</i>	cf. <i>giv'a</i>
hill.F.SG-CS	the-iris-PL	hill.F.SG
(hill name)		‘hill’

(7) Non-toponymic

a. <i>bat.ey</i>	<i>ha-morá</i>	cf. <i>bat.ím</i>
house.PL.CS	teacher.F.SG	house.PL
‘teachers’ houses’ (compositional)		‘houses’

b. <i>bat.ey</i>	<i>sefér</i>
house.PL.CS	book
‘school’ (non-compositional)	

(8) Toponymic

a. <i>gan.ey</i>	<i>tikvá</i>	cf. <i>gan-ím</i>
garden.PL.CS	hope	garden-PL
(city name)		‘gardens’

b. <i>nican.ey</i>	<i>‘óz</i>	cf. <i>nican-ím</i>
bud.PL.CS	strength	bud-PL
(village name)		‘buds’

Having established the morpho-phonological similarities between toponymic and non-toponymic Constructs in the language, we now turn to the diagnostics that distinguish them.

2.2. Semantic headedness

Applying the so-called ‘IS A’ condition of Allan (1978), adapted here as in (9), Borer (2013) observes that Constructs fall into two major subclasses: (i) compositional (i.e., R- and M-Constructs) and (ii) non-compositional (i.e., compound Constructs). Specifically, while R- and M-constructs have a compositional meaning in that the head denotes a modified version of its free-state counterpart (e.g., 10), compound Constructs lack such a reading (e.g., 11) and are instead associated with syntactic opacity, being best understood as non-compositional.¹¹

(9) [N1+N2] is a N1 = TRUE/FALSE

(10) Compositional:

a. [*beyt* *mora*] IS A *bayit* = TRUE
house.CS teacher IS A house
‘A teacher’s house...’ (R-Constructs)

b. [*xalon* *zxuxit*] IS A *xalon* = TRUE
window.CS glass IS A window
‘a glass window...’ (M-Constructs)

(11) Non-compositional:

a. [*beyt* *sefer*] IS A *bayit* = FALSE
house.CS book IS A house
‘School’ (lit. book house; compound Construct)

b. [*yošev* *roš*] IS A *yošev* = FALSE
sitter.CS head IS A sitter
‘Chairman’ (lit. head sitter; compound Construct)

Adopting the same test for toponymic Constructs allows us to determine whether they pattern with either compositional or non-compositional Constructs in the language. As the following

¹¹ Examples (10)-(11) are adapted from Borer (2013: 218)

examples show, some toponyms have a compositional meaning, in that the denotatum of the head is a modified version of its free-state counterpart (e.g., (12)). In contrast, others lack such reading (e.g., (13)), motivating the classification of the former as compositional Constructs and the latter as non-compositional Constructs (i.e., compound Constructs).

- (12) a. [*har ha-zeytim*] IS A *har* = TRUE
 mountain.CS the-olive.PL IS A mountain
 (mountain name)
- b. [*gva'ot ha-šokolad*] ARE *gv'a'ot* = TRUE
 hill.CS.PL the-chocolate ARE hill.PL
 (hills' name)¹²

- (13) a. [*tel aviv*] IS A *tel* = FALSE
 mound.CS spring IS A mound
 (city name)
- b. [*'ir yam-im*] IS A *'ir* = FALSE
 city.CS sea-PL IS A city
 (neighborhood name)

Thus, we conclude that toponymic Constructs fall into two distinct subclasses based on the semantic headedness test. Some toponymic Constructs have a compositional meaning and are syntactically transparent, correlating with compositional Constructs (cf. 10-12), whereas others lack such reading and are syntactically opaque, correlating with non-compositional Constructs (cf. 11-13).

¹² Prescriptively, the construct state form of the plural noun *giv'a* 'hill' is *giv'ot*, while the corresponding free-state form is *gva'ot*. In colloquial Hebrew, however, the form *gva'ot* is often used in Constructs.

Note that this test only shows that some toponymic Constructs may be classified as Compound Constructs (e.g., (13a–b)). It does not distinguish between the two subclasses of compositional Constructs, namely R-Constructs vs. M-Constructs. We therefore turn to the test of pronominal anaphora, which exclusively distinguishes between these two classes. As will be shown, toponymic Constructs with compositional meaning (e.g., (12a–b)) fall under the class of M-Constructs rather than that of R-Constructs.

2.3. Pronominal anaphora

Given that some toponymic Constructs fall within the class of compositional Constructs (i.e., R- and M-Constructs), we now determine which subclass they belong to by applying the test of pronominal anaphora. Borer (2013) demonstrates that this test distinguishes M-constructs from both R-constructs and (non-compositional) compound Constructs along two factors: (a) pronominal reference to the head and (b) pronominal reference to the non-head.

Specifically, M-constructs, like R-constructs, allow pronominal reference to the head, a possibility not available for compound constructs (cf. 14-16). However, M-constructs, like compound Constructs, disallow pronominal reference to the non-head, whereas such reference is fully licit with non-heads of R-constructs (cf. 14-15).¹³

(14) R-constructs:

- a. [*beyt (ha-)mora_i*] *ve-xacer-a_i*
House.CS (the-)teacher_i and-yard-her_i
'The/a teacher_i's house and her_i yard'
- b. '*amdu šam šney [batey_i [morot mi-xul]] ve-'exad_i šel mora mi-be'er ševa*
stood there two houses_i teachers from-abroad and one_i of teacher from-Be'er Sheba
'There were two houses there of teachers from abroad, and one of a teacher from Be'er Sheba'

¹³ Examples (14-16) are adapted from Borer (2011: 495-505).

(15) M-constructs:

- a. *[*xalon (ha-)zxuxit_i*] *ve-dalt-a_i*
 window (the-)glass_i and-door-her_i
 Intended: ‘the/a glass_i window and its_i door’
- b. *kaniti šney [xalonot_i zxuxit] ve-‘exad_i mi-plastik*
 bought two window.PL.M_i glass.F and-one.M_i from-plastic
 ‘I bought two glass windows and one from plastic’

(16) Compound Constructs:

- a. *[*beyt (ha-)xolim_i*] *ve-mitat-am_i*
 house (the-)patients_i and bed-theirs_i
 Intended: ‘The patients’_i home (=hospital) and their_i bed’
- b. **hu makir šney [batey_i xolim] ve-‘exad_i le-yetomim* cf. *beyt yetomim*
 he know two houses_i patients and-one_i for-orphans house orphans
 Intended: ‘He knows two hospitals and orphanage’ ‘orphanage’

Applying the anaphoric test to compositional toponymic Constructs, such toponyms are predicted to be classified as R-Constructs if they allow pronominal reference to both the head and the non-head. In contrast, they are predicted to be classified as M-Constructs if they allow pronominal reference only to the head. As illustrated in (17), only the latter prediction is borne out, suggesting that compositional toponymic Constructs are in actuality M-Constructs. More concretely, in (17a) the singular pronominal suffix ‘o cannot be used to refer to the non-head of the Construct *ha-šokolad*. However, in (17b) the demonstrative *ha-‘elu* can refer exclusively to the head of the Construct *gva’ot*.

(17) Compositional toponymic Constructs:

- a. *[*gva’ot ha-šokolad_i*] *ve-civ-‘o_i*
 hill.PL the-chocolate_i and-color-its_i
 Intended: ‘Chocolate_i Hills and its_i color’

b. [gva'ot_i ha-šokolad] ba-filipinim yoter yafoṭ me-ha-'elu_i
 hill.CS.PL_i the-chocolate_i in+the-Philippines more beautiful from-the-these_i
mi-kurkar be-nes ciyona
 from-sandstone in Nes-Ziona
 'Chocolate Hills in the Philippines are more beautiful than those of sandstone in
 Ness-Ziona.'

cf. gva'ot ha-karkur
 hill.CS.PL the-sandstone
 'Sandstone Hills (in Ness-Ziona)'

Just like Compound Constructs in (16), we further observe from (18) that non-compositional toponymic Constructs do not allow either the head (e.g., (18a)) or the non-head (e.g., (18b)) to function as antecedents for pronominal reference. This finding further supports their classification as compound Constructs.

(18) Non-compositional toponymic Constructs:

a. *[tel aviv] ve-ha-temperaturoṭ ha-'ofyaniyot lo_i
 mound.CS spring_i and-the-temperatures the-typical to+it_i
 Intended: 'Tel Aviv_i and its_i typical temperatures'

b. *[tel_i aviv] yoter gadol mi-ze_i šel Megiddo cf. Tel Megiddo
 mound.CS_i spring more big from-this_i of Megiddo mound.CS Megiddo
 Intended: 'Tel_i Aviv is bigger than that_i of Megiddo' (toponym)

To sum up, the anaphora test provides evidence for classifying compositional toponymic Constructs as M-Constructs, to the exclusion of R-Constructs, and non-compositional toponymic Constructs as compound Constructs. One may still wonder whether toponymic Constructs that instantiate the class of R-Constructs exist. As the next test of non-head modification demonstrates (§2.4), the answer to this question is negative.

2.4. Non-head modification

The diagnostics considered thus far suggest that toponymic Constructs share core properties with M-Constructs and Compound Constructs, to the exclusion of R-Constructs. This raises the question of whether any toponymic Constructs exhibit properties characteristic of R-Constructs. The goal of this section is therefore to examine this possibility by applying the diagnostic of non-head modification. Recall from Table 1 that modification of the non-head is illicit in both M-Constructs and Compound Constructs but fully permitted in R-Constructs. It is therefore predicted that if (some) toponymic Constructs allow modification of their non-head, they are expected to qualify as toponymic R-Constructs. As shown below, however, this prediction is not borne out.

Specifically, as observed by Borer (2013), the (im)possibility of modifying non-heads distinguishes R-constructs from both M-constructs and compound constructs. For example, non-heads of R-constructs can be freely pluralized (e.g., (19a)), modified by cardinals or quantifiers (e.g., (20a)), or by determiners and reference-denoting adjectives (e.g., (21a)). Such modifications do not hold for non-heads of M-constructs or compound constructs (cf. b-c in (19)-(21)). Unlike free pluralization, we further observe that plural-marked non-heads in M-constructs are interpreted as property modifiers (cf. (19bii)), whereas in compound constructs they are inherent to the associated non-compositional meaning (cf. (19c)).¹⁴

(19) a. R-Construct:

beyt (*ha-*)*morot*

house.CS (the-)teacher.PL

‘(the) teachers’ house’

b. M-Construct:

(i) **xalon* (*ha-*)*zxuxiyot*

window.CS (the-)glass.PL

Intended: ‘(the/a) (multiple-)glasses’ window’

¹⁴ Examples (19)-(21) are adapted from Borer (2013: 214-230).

(ii) *madaḥ* (ha-)sḥar.im cf. **madaḥ* (ha-)seḥer
 shelf.CS (the-)book.PL shelf.CS (the-)book.SG
 ‘(the) bookshelf’ Intended: ‘(the) bookshelf’

c. Compound Construct:

beyt (ha-)xolim cf. **beyt* (ha-)xole
 house.CS (the-)patients house.CS (the-)patient
 ‘(the) hospital’ Intended: ‘(the) hospital’

(20) a. R-Construct:

beyt {šalos} (ha-)morot; *beyt* {kol} mora
 house.CS three (the-)teachers; house.CS every teacher
 ‘(the) house of three/many teachers’; ‘every teacher's house’

b. M-Construct:

(i) *madaḥ* { *me’a/*harbe } sḥarim;
 shelf.CS hundred/many books;
 Intended: ‘a shelf of hundred/many books’

(ii) *xalon* { *kol } zxuxit
 window.CS every glass
 Intended: ‘a window from every (type of) glass’

c. Compound Construct:

beyt { *me’a/*harbe } xolim;
 house.CS hundred/many patients
 Intended: ‘a hospital of hundred/many patients’

(21) a. R-Construct:

beyt { *’eyzo } mora { *kolšehi } /*mesuyemet}
 house.M.SG.CS some.F.SG teacher.F.SG some.F.SG / specific.F.SG

‘a house of some/specific teacher’

b. M-Construct:

xalon {**eyzo*} *zxuxit* {**kolšehi* / **mesuyemet*}
window.M.SG.CS some.F.SG glass.F.SG some.F.SG / specific.F.SG
Intended: ‘a window of some/specific glass’

c. Compound Construct:

beyt {**eyzešehem*} *xolim* {**kolšehem* / **mesuyamim*}
house.M.SG.CS some.M.PL patient.M.PL some.M.PL / specific.M.PL
Intended: ‘a hospital of some/specific patients’

If some toponymic Constructs are indeed subclasses of R-Constructs, it is predicted that their non-heads should be accessible to modification, just like those of their non-toponymic counterparts. As shown in (22)–(24), however, the data do not support this prediction. Instead, such toponyms display precisely the same modificational restrictions of the non-head as expected of both Compound Constructs and M-Constructs. That is, free pluralization is impossible unless the plural itself is interpreted either as a property modifier, if the toponym is an M-construct (e.g., (22a)), or as a conventionalized subpart, if it is a compound construct (e.g., (22b)). In (23), we further observe that their non-heads cannot be modified by cardinals or quantifiers, nor by determiners and reference-denoting adjectives (e.g., 24).¹⁵

(22) a. Toponymic M-Constructs:

(i) <i>har</i>	<i>ha-zeytim</i> ;	cf. <i>*har</i>	<i>ha-zayit</i>
mountain.CS	the-olive.PL	mountain.CS	the-olive.SG
‘The olive mountain’		Intended: ‘The olive mountain’	(in reference to
(mountain name)			a mountain on which olive trees grow)

¹⁵ One might question the relevance of modifying the non-head, given that toponyms are proper names and may therefore be viewed as fixed expressions. However, such toponyms can also be used as common noun phrases. The contrasts observed here thus provide indirect evidence for their classification (i.e., as M-Constructs, R-Constructs, or Compound Constructs). Also, note that proper names are not necessarily fixed expressions; on the syntactic flexibility of proper names, see, for example, Matushansky (2015).

- | | |
|--|--|
| (ii) <i>gva'ot ha-šokolad</i> ;
hills.CS the chocolate.SG
'The chocolate hills'
(hills' name) | cf. <i>*gva'ot ha-šokoladim</i>
hills.CS the chocolate.PL
Intended: 'The chocolate hills' (in reference to
hills that may be identified by the
concept of chocolate) |
|--|--|

b. Toponymic Compound Constructs:

- | | |
|--|--|
| (i) <i>'ir yam-im</i> ;
city.CS sea-PL
'City of seas'
(neighborhood name) | cf. <i>*'ir yam</i>
city.CS sea.SG
(Intended to refer to a city that is identified by
the concept of sea) |
|--|--|

- | | |
|---|---|
| (ii) <i>petax tikva</i> ;
opening.CS hope.SG
'Opening of hope'
(city name) | cf. <i>*petax tikvot</i>
opening.CS hope.PL
Intended: 'Opening of hope' |
|---|---|

(23) a. Toponymic M-Constructs:

- (i) *har {*me'a/*harbe} ha-zeytim*
 mountain.CS hundred/many the-olive.PL
 Intended: 'The mountain of hundred/many olives'
- (ii) *gva'ot {*rov} ha-šokolad*
 hills.CS most the-chocolate
 Intended: 'The hills of the most (quantity) chocolate'

b. Toponymic Compound Constructs:

- (i) *'ir {*me'a/*harbe} yam-im*
 city.CS hundred/many sea-PL
 Intended: 'The city of hundred/many seas'

(ii) *petax* *{*kol} tikva*
 opening.CS every hope
 Intended: ‘Opening of every hope’

(24) a. Toponymic M-Constructs:

har *{*eizešehem} zeytim* *{*kolšehem/*mesuyamim}*
 mountain.SG.CS some.PL olive.PL some.PL / specific.PL
 Intended: ‘a hill of some/specific irises’

b. Toponymic Compound Constructs:

’ir *{*eizešehem} yam-im* *{*kolšehem/*mesuyamim}*
 city.SG.CS some.PL sea-PL some.PL / specific.PL
 Intended: ‘a city of some/specific seas’

These data further support the conclusion that toponymic Constructs comprise two major subclasses (as summarized in Table 1): (a) toponymic Compound Constructs and (b) toponymic M-Constructs, to the exclusion of a third class of toponymic R-Constructs.¹⁶

2.5. Wordhood vs. phrasal status

Having established that toponymic Constructs instantiate Compound Constructs and M-Constructs (but not R-Constructs), the goal of this section is to determine the grammatical status of each subclass on the basis of its external syntax, that is, whether it patterns as a word or as a phrasal constituent. Specifically, it is well known that compounds display properties characteristic of words (however defined); for example, they are inflected as a whole (see, e.g.,

¹⁶ Note that cross-linguistically toponymic N+N combinations are not unique to Semitic as illustrated by the following examples from genetically unrelated languages:

(i) French: <i>Bois-le-Roi</i>	(ii) German: <i>König-s-berg</i>	(iii) Tamil: <i>yānai-malai</i>
wood-the-king	king-LM-mountain	elephant-hill
(Commune name)	(City name)	(Hill name)

Whether such toponyms may be classified as Constructs of some sort is a question I leave open. I thank Madhusmitha Venkatesan for bringing the Tamil example to my attention.

Lieber & Štekauer 2009; De Belder 2022). M-Constructs, by contrast, exhibit properties characteristic of phrasal constituents. As shown in §§2.2–2.3, their head remains semantically transparent and may function as the antecedent of pronominal anaphora, suggesting that M-Constructs have phrasal status.

Importantly, if toponymic Constructs genuinely instantiate Compound Constructs and M-Constructs, it is further predicted that the former should have wordhood status, whereas the latter should have phrasal status. As illustrated below, this prediction is borne out.

Consider, for example, the paradigms in (29) and (30), illustrating toponymic Compound Constructs and toponymic M-Constructs, respectively. Toponymic Compound Constructs inflect as a single unit, as shown in (29a–b). Specifically, in (29a), neither nominal subpart of the compound determines the inflection of the copula: the copula is inflected for feminine gender, whereas both the head and the non-head are masculine. We further observe in (29b) that toponymic Compound Constructs may undergo derivational affixation as a unit, for instance, through the merger of the adjectivizing suffix *-i*.

(29) Toponymic Compound Constructs

- a. [*tel* *aviv*] *hi* *ha-miflat ha-tov* *be-yoter* *la-nefeš*
 mound.M.SG.CS spring.M.SG be.F.SG the-refuge the-good in-more to+the-soul
 ‘Tel Aviv is the best refuge for the soul.’¹⁷

- b. ‘*atem ha-zug haxi* [*tel* *aviv*] *-i*] *še-yeš*
 you.PL the-couple most [mound.CS spring]-ADJ] that-there is
 ‘You are the most Tel-Aviv-like couple there is.’¹⁸

In contrast, toponymic M-Constructs behave as full phrasal constituents. Specifically, (30a) demonstrates that only the ϕ -features of the head determine the inflection of the copula. It is further shown in (30b) that the adjectivizing suffix *-i* reduces the acceptability of the toponymic M-Construct.

¹⁷ Example retrieved from [LINK](#)

¹⁸ Example retrieved from [LINK](#)

(30) Toponymic M-constructs

- a. [gva 'ot ha-šokolad] hen { *hem / *hu / *hi }
hill.F.PL.CS the-chocolate.M.SG be.F.PL be.M.PL / be.M.SG be.F.SG
tofa 'at teva yixudit
phenomenon nature unique
'Chocolate Hills are a unique natural phenomenon.'¹⁹

- b. ze ha-nof haxi ??[gva 'ot šokolad] -i] še-ani makir be-amerika
this the-landscape most [hill.CS.PL chocolate]-ADJ that-I know.PRS.SG in-America
'This is the most Chocolate-Hills-like landscape that I know in America.'²⁰

These distinctions lead to the conclusion that toponymic Compound Constructs have wordhood status, whereas toponymic M-Constructs have phrasal status.

3. Assigning structure to toponymic Constructs

The diagnostics discussed in the previous sections show that toponymic Constructs fall into two distinct subclasses, each characterized by different grammatical properties: (a) toponymic Compound Constructs and (b) toponymic M-Constructs. The grammatical differences established by the diagnostics in §2 suggest that these subclasses correspond to distinct underlying structures. In what follows, I propose a formal analysis that derives these structures. Specifically, by structural parsimony, it is argued that the structural configurations of toponymic Constructs are reducible to structures that are independently available in the language for non-toponymic Constructs. §3.1 discusses the structure of toponymic M-Constructs and §3.2 discusses the structure of toponymic Compound Constructs.

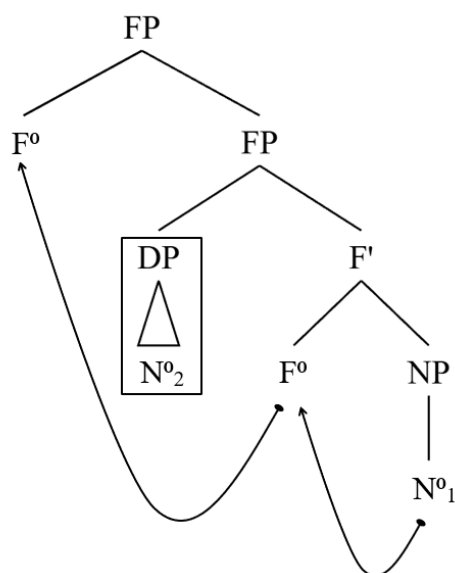
¹⁹ Example retrieved from [LINK](#)

²⁰ The acceptability of this example may vary across speakers. As a native speaker of Hebrew, I judge (29b) to be considerably more natural than (30b), and it is this contrast that forms the basis of the conclusion reached in this section. A more systematic investigation of these judgments is left for future research.

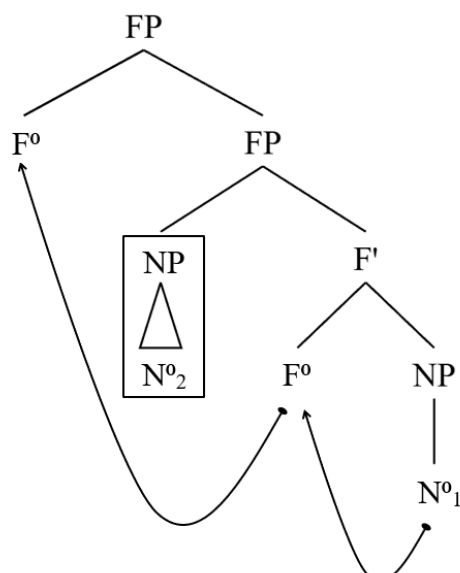
3.1. Internal syntax of toponymic M-constructs

The structures that Borer (2013: 231) proposes for (non-toponymic) R-constructs and M-constructs are given in (25a) and (25b), respectively. In both structures, the non-head always merges in the specifier position of some nominal functional projection FP associated with the head N_1 .²¹ Both differ, however, insofar as in R-constructs the non-head is a referential constituent, a full DP, whereas in M-constructs it is a property modifier, a predicate NP. Following standard assumptions on the syntactic derivation of Hebrew construct nominals (e.g., Ritter 1991), the head N_1 raises to a higher functional head position above its specifier, the non-head N_2 , deriving the attested surface order $N_1(\text{head}) >> N_2(\text{non-head})$.

(25) a. R-Construct:



b. M-Construct:



Adopting these assumptions, and in the interest of structural parsimony, I propose that toponymic Constructs that have the core properties of M-constructs reduce to the structure

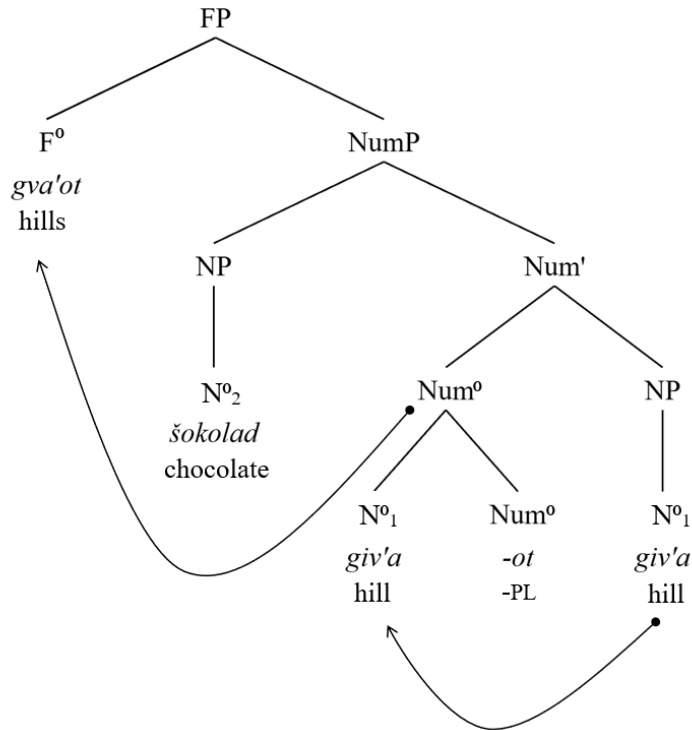
²¹ Note that this assumption differs from Ritter (1991) and subsequent work (see also Preminger 2020), where the non-head is assumed to be base-generated within the lexical projection NP of the head N_1 . For present purposes, however, this technical difference in implementation has no crucial consequences for the preliminary analysis developed here.

independently proposed for M-Constructs in the language. As illustrated in (26), toponymic M-Constructs are assigned the structure of a phrasal constituent, consistent with their phrasal status as established by the diagnostics in §2. Specifically, they are composed of a non-head, N₂, corresponding to a property-denoting modifier, an NP predicate (i.e., *chocolate* in (26)).²² As in non-toponymic M-constructs, the non-head merges directly in the specifier of some nominal functional projection associated with the head N⁰₁ (i.e., *hill* in (26)). For present purposes, I take this projection to be NumP (Ritter 1991), encoding the number specification of the head. Accordingly, the head raises to Num⁰, where it acquires number specification (i.e., the plural suffix *-ot* in (26)), and then to a higher functional head above its specifier (i.e., the non-head *chocolate*), deriving the attested order (e.g., N₁ hills (head) >> N₂ chocolate (non-head) in (26)).²³

²² Unlike R-constructs, the definite article *ha-* on the non-head in M-constructs does not render it referential (i.e., DP). Rather, definiteness is interpreted on the head; see Rothstein (2012). The same occurs in toponymic M-constructs, which are typically used as definite descriptions, e.g., *gva'ot ha-sokolad* lit: ‘the chocolate hills’ referring to specific hills characterized by a non-specific kind of chocolate. I suggest that this pattern reflect the post-syntactic PF operation known as Local-Dislocation (Embick & Noyer 2001), whereby the definite article is overtly realized on the non-head but semantically interpreted on the head of the Construct. Given the fact that this operation takes place post-syntactically, I do not realize the definite article *ha-* on the NP predicate *chocolate* in the syntactic representation in (26). A full account of this is beyond the scope of this paper.

²³ Borer (2013: 231) remains silent on which specific specifier hosts the non-head in either M-constructs or R-constructs. It is therefore only for expositional purposes that I adopt Ritter’s (1991) NumP, though the structures proposed here for N+N toponyms are compatible with the XS-model.

(26) *Chocolate Hills* as an M-Construct in Hebrew:



As for their meaning, recall that M-constructs have compositional meaning; they involve the semantics of property modification. According to Borer (2013: 232), non-heads of M-constructs are unsaturated NP predicates. Once such NP predicates merge with the head of M-constructs, N^0_1 , they are interpreted as (saturated) property-denoting predicate modifiers.

Having reduced some toponymic Constructs to the class of M-constructs, it is reasonable to assume that the same interpretive process is in principle available to them. In toponymic M-constructs, the non-head is thus an unsaturated NP predicate. Only by virtue of its merger with the head N^0_1 , it is interpreted as a saturated predicate functioning as a property modifier. This, in turn, gives rise to a compositional meaning. For (26), *Chocolate Hills*, this can be roughly paraphrased as ‘hills that are (somewhat) related to or identified by chocolate’. Indeed, as documented (e.g., Salomon 2011), it is the chocolate-brown color of the hills’ grass that gave rise to the name and is what language users associate with its denotatum.

3.2. Internal syntax of toponymic compound Constructs

As established by the diagnostics in §2, toponymic Compound Constructs are non-compositional and have wordhood status. This contrasts sharply with toponymic M-Constructs, which are compositional and have the status of full phrasal constituents. This contrast raises the question of how Hebrew compounds are assigned their associated non-compositional meaning and whether this follows from any structural considerations. One approach to addressing this question, which I adopt here, is provided by the XS-model (Borer 2005, 2013, 2017). Within this model, differences in meaning are attributed to differences in syntactic structure. More explicitly, structurally complex words and phrases with compositional meanings are associated with one type of syntactic structure, whereas those with non-compositional meanings are associated with a different type (Borer 2017).

Before proceeding, a brief clarification of what is meant by non-compositional meaning here is in order. In the XS-model, non-compositional meanings of words and phrases are taken to correspond to atomic Content units. The term Content (akin to Fregean Sense) refers to conceptual information stored in the Encyclopedia, a repository of world knowledge (Borer 2013a, 2014). Syntactic constituents do not intrinsically possess Content; rather, they are matched with Content via encyclopedic search - *en-search* - an operation that links linguistic form with meaning. A successful single *en-search* returns a single atomic Content unit for a qualifying linguistic domain (Borer 2013b: 238–239).

Borer (2013: 239; 2014: 85) defines the qualifying linguistic domain structurally as follows (simplified): a structural configuration containing at least one functional projection cannot return a single atomic Content unit and must therefore be interpreted compositionally. In the absence of functional structure, however, a single *en-search* may apply successfully and return an atomic Content unit for the relevant domain.

For the analysis developed here, this mechanism is relevant for deriving the non-compositional meaning associated with toponymic Compound Constructs. To illustrate how this mechanism operates, I first consider Borer's (2013) analysis of non-toponymic Compound Constructs.

Specifically, Borer (2013) proposes that Hebrew compound Constructs are associated with a distinct syntactic structure that allows their non-compositional meaning to emerge. According to this proposal, the internal structure of Hebrew compounds involves syntactic incorporation for non-compositional Content to emerge. Assuming that no grammatical principles are violated, under this view, Hebrew compound Constructs are derived from M-constructs that have undergone syntactic incorporation of the head of the non-head, N_2 , into the raised head N_1 .²⁴ Prior to incorporation, the boxed domain in (27a) cannot be an input to a single en-search due to the presence of intervening functional material between the head and the non-head. After incorporation, however, the resulting structure no longer contains intervening functional material. It therefore forms the boxed domain in (27b), a branching head, which serves as a possible input to a single en-search matching the relevant atomic Content to that domain.²⁵

- (27) a. $[_{FP1} \boxed{N_1 [_{FP1} [_{NP} N_2]} N_1 [_{NP} N_1]}]$
 b. $[_{FP1} \boxed{N_1 + N_2} [_{FP1} [_{NP} N_2]} N_1 [_{NP} N_1]}]$
 beyt sefer *sefer* *beyt* *beyt*
 SCHOOL (book) (house)

Adopting these assumptions, it follows then that toponymic compounds must correspond to branching heads to be assigned non-compositional Content. But what could essentially be the non-compositional Content that toponyms are assigned in this case? I propose that it is precisely those concepts associated with geo-ontological categories, broadly corresponding to categories such as [CITY], [VILLAGE], [HILL], etc. (cf. Smith & Mark 2001) that are then matched to the

²⁴ See Borer (2013) for arguments that Hebrew Compound Constructs are derived exclusively from the structure of M-Constructs. Also, Borer's conception of syntactic incorporation in this case differs from that of Baker (1988); see Borer (2013: 236) for details.

²⁵ On the assumption that copies are not assigned meaning by the Encyclopedia; that is, only syntactically annotated PF representations are accessible to encyclopedic interpretation; see Borer (2011). Note further that while non-compositional Content in Hebrew compound Constructs is contingent on incorporation, in the absence of incorporation their Content must remain compositional; cross-linguistically, however, incorporated structures in principle allow compositional readings. See Borer (2013: 241) for discussion.

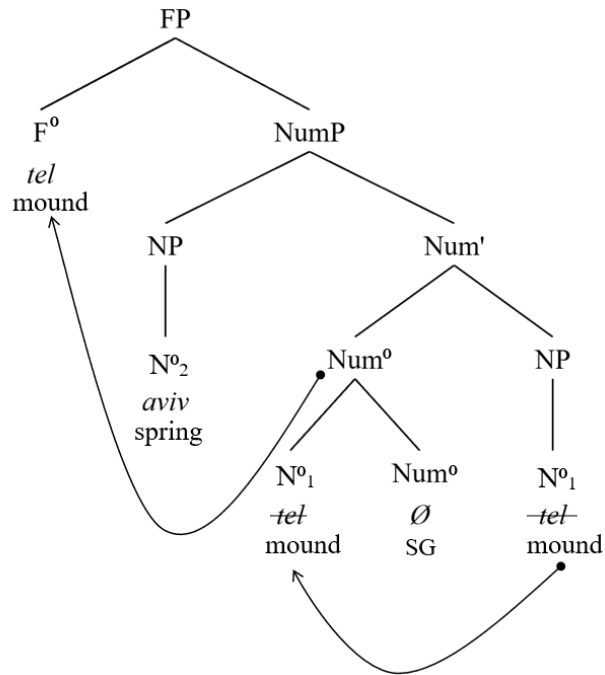
qualifying linguistic domain (i.e., the branching head) by means of a successful single en-search application.²⁶

Accordingly, I assume the following syntactic derivation: as with non-toponymic compound Constructs, I take toponymic compounds to be derived from M-constructs via incorporation of the head of the non-head into the raised head, forming a branching head. This, in turn, results in the boxed domain in (28b), which is a possible input to a single en-search that matches the relevant non-compositional Content of Tel-Aviv (e.g., [CITY]) to that domain.²⁷ This mapping process is represented in (28b) by the dashed arrows. Importantly, prior to incorporation, the structure in (28a) is not a possible input to a single en-search, as it contains intervening functional material between the head and the non-head (i.e., [NumP]) and can therefore be interpreted only compositionally.

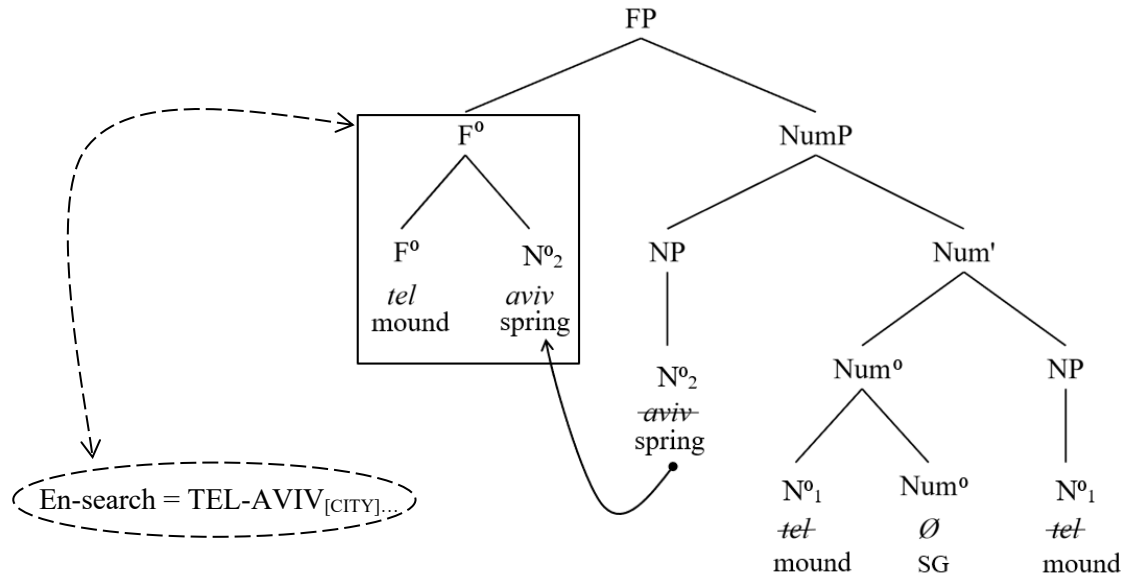
²⁶ Clearly, this is a simplified account. It is very plausible that a single toponym is associated with a (structured) set of geo-ontological categories.

²⁷ A question that arises is how a toponym acquires the proper name status, viz., what determines its “namehood”? I do not attempt to address this issue here. There are competing accounts of the nature of proper names, mostly developed for personal names, each with different assumptions and predictions (cf. Anderson 2007; Borer 2005; Ghomeshi & Massam 2009; Jambrović 2024; Longobardi 1994; Matushansky 2006; Ursini & Zhang 2023). Determining which account best captures proper-namehood in toponyms falls beyond the scope of this paper.

(28) a. *Tel-Aviv* as an M-Construct in Hebrew:



b. *Tel-Aviv* as a compound Construct in Hebrew:



It is important to note that, prior to incorporation, N+N toponyms are de facto M-constructs associated with compositional meaning, as in (28a). This compositional meaning can be understood as capturing the naming convention underlying the toponym. For example, in *Tel Aviv*, this yields a compositional reading in which *aviv* ‘spring’, the non-head, is a property-

denoting NP modifier of the head *tel* ‘mound’, thereby capturing features of the geographical environment, culture, and/or history underlying the place name. That toponymic compounds originate as M-constructs is especially evident in cases where language users reason about why a place bears a particular name: the pre-incorporation M-construct thus represents exactly that compositional meaning that is, in principle, available in the grammar.

Having reduced the internal structure of toponymic Constructs in Hebrew to that of M-Constructs and Compound Constructs, it is concluded that toponymic Constructs do not constitute a uniform class. Rather, they instantiate two subclasses: (i) toponymic Compound Constructs and (ii) toponymic M-Constructs, with the structure of the former derived from that of the latter.

4. Concluding remarks

The goal of this paper was to advance the claim that toponyms are structurally decomposable. Evidence for this claim came from Hebrew toponymic Constructs displaying noun–noun concatenation. While previous research has primarily focused on the distribution of such toponyms, the present study investigated their internal grammatical structure and their status within the broader typology of Hebrew Constructs. Applying a range of diagnostics, it was shown that toponymic Constructs do not constitute a uniform class. Rather, they instantiate only two subclasses available in the language: (i) toponymic Compound Constructs and (ii) toponymic M-Constructs, to the exclusion of toponymic R-Constructs.

This typological classification in turn motivated the formal analysis developed in §3. Specifically, the fact that toponymic Constructs systematically pattern as two distinct grammatical classes suggests that they are structurally decomposable in two distinct ways. By structural parsimony, I argued that the two subclasses reduce to independently motivated structures already available for non-toponymic Constructs in the language. Toponymic M-Constructs were analyzed as compositional phrasal constituents, whereas toponymic Compound Constructs having wordhood status were analyzed as incorporated structures associated with non-compositional meaning. It was further proposed that the non-compositional meaning

associated with toponymic Compound Constructs corresponds to atomic Content units linked to geo-ontological categories (e.g., [CITY], [VILLAGE], [HILL]; cf. Smith & Mark 2001). The analysis thus suggests a principled mapping between geographical concepts and syntactic structure.

Several questions remain open for future research. First, the fact that N+N toponyms do not share key properties with R-constructs raises the question of why this is the case and calls for a explanation; it may be that the absence of the properties of R-constructs are the ones that characterize toponyms more generally. Second, it remains to be examined to what extent N+N toponyms are attested cross-linguistically, and whether the reductionist approach adopted here extends beyond Hebrew. Finally, a puzzling asymmetry emerges between toponymic M-constructs and toponymic compound Constructs: the former typically denote geographically natural entities (e.g., hills, seas), whereas the latter tend to refer to human-populated settlements (e.g., cities, villages); an asymmetry that appears to mirror Van Langendonck's (2007) hierarchy for proper names more generally. Whether this generalization holds in a systematic way, and what its broader implications are, requires further cross-linguistic investigations.

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Appendix: not applicable

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